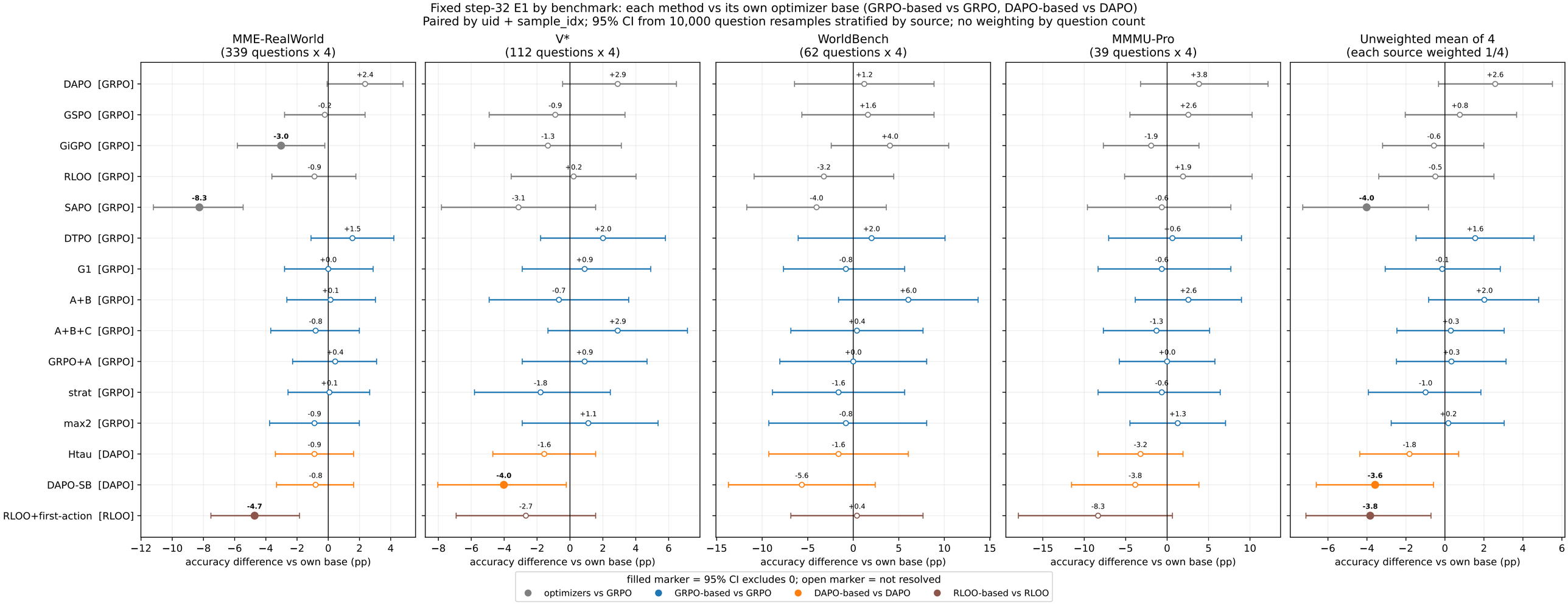
Fixed step-32 E1 by benchmark: each method vs its own optimizer base (GRPO-based vs GRPO, DAPO-based vs DAPO)
Paired by uid + sample_idx; 95% CI from 10,000 question resamples stratified by source; no weighting by question count
MME-RealWorld
(339 questions x 4)
V*
(112 questions x 4)
WorldBench
(62 questions x 4)
MMMU-Pro
(39 questions x 4)
Unweighted mean of 4
(each source weighted 1/4)
DAPO [GRPO]
GSPO [GRPO]
GiGPO [GRPO]
RLOO [GRPO]
SAPO [GRPO]
DTPO [GRPO]
G1 [GRPO]
A+B [GRPO]
A+B+C [GRPO]
GRPO+A [GRPO]
strat [GRPO]
max2 [GRPO]
Htau [DAPO]
DAPO-SB [DAPO]
RLOO+first-action [RLOO]
+2.4 -0.2 -3.0 -0.9 -8.3 +1.5 +0.0 +0.1 -0.8 +0.4 +0.1 -0.9 -0.9 -0.8 -4.7
+2.9 -0.9 -1.3 +0.2 -3.1 +2.0 +0.9 -0.7 +2.9 +0.9 -1.8 +1.1 -1.6 -4.0 -2.7
+1.2 +1.6 +4.0 -3.2 -4.0 +2.0 -0.8 +6.0 +0.4 +0.0 -1.6 -0.8 -1.6 -5.6 +0.4
+3.8 +2.6 -1.9 +1.9 -0.6 -0.6 -0.6 +2.6 -1.3 +0.0 -0.6 +1.3 -3.2 -3.8 -8.3
+2.6 +0.8 -0.6 -0.5 -4.0 +1.6 -0.1 +2.0 +0.3 +0.3 -1.0 +0.2 -1.8 -3.6 -3.8
−12 −10 −8 −6 −4 −2 0 2 4
−8 −6 −4 −2 0 2 4 6
−15 −10 −5 0 5 10 15
−15 −10 −5 0 5 10
−6 −4 −2 0 2 4 6
accuracy difference vs own base (pp)
filled marker = 95% CI excludes 0; open marker = not resolved
optimizers vs GRPO
GRPO-based vs GRPO
DAPO-based vs DAPO
RLOO-based vs RLOO